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CONNECTOR WITH TETHERED CAPS

Abstract

A submersible cable connector for connecting a plurality of cable ends is provided. The submersible cable connector includes a connector body and an encapsulation member covering the connector body. The connector body has one or more cable receiving apertures or cavities and one or more fastener receiving apertures in communication with the cable receiving apertures. The encapsulation member includes integrally molded tubular cable ports in communication with the cable receiving apertures, spaced apart tubular fastener ports in communication with the fastener receiving apertures, and tethered sealing cap assemblies monolithically or integrally formed into the encapsulation member, and each configured to seal respective one of the fastener ports.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of co-pending U.S. application Ser. No. 17/894,405 filed Aug. 24, 2022, which is based on and claims benefit from U.S. Provisional Patent Application No. 63/390,225 filed on Jul. 18, 2022 and from U.S. Provisional Application No. 63/237,305 filed on Aug. 26, 2021, the contents of each are incorporated herein in their entirety by reference.

BACKGROUND

Field

[0002] The present disclosure relates generally to electrical connectors, and more particularly, to submersible electrical cable connectors for connecting a plurality of cable ends.

Description of the Related Art

[0003] Underground and/or submersible electrical cable connectors are generally well-known and widely used for making connections between power conductors in electrical power distribution systems. These connectors typically include a body formed of a rigid material and having at least a few sets of spaced apart ports extending from the body at right angle relative one another. Generally, one of the sets is adapted to receive the ends of electrical conductors, and another set is adapted to receive fasteners for securing the ends of the electrical conductors in a mechanical and/or electrical engagement to the body. To ensure longevity of the connectors in an underground environment and/or to electrically insulate them, these connectors are typically coated in an insulating covering. Additionally, various auxiliary components are provided with these connectors, such as sealing plugs for capping the set of ports that receives fasteners and cable adapters that are insertable into the set of ports that receives the electrical conductors to accommodate conductors of various sizes. Unfortunately, however, when installing and/or maintaining the connectors in the field, utility workers often have trouble removing these sealing plugs. It is also common for the utility workers to drop, misplace or lose these sealing plugs when they are removed. Prior solutions adopted by connector manufacturers to provide sealing plugs that mate with the body of these connectors adds to the assembly time and cost to manufacture the connectors.

SUMMARY

[0004] The present disclosure provides embodiments of a submersible electrical cable connector that, among other benefits, has cap assemblies that are monolithically integrated with an encapsulation overmold that encases the rigid, electrically conductive body of the cable connector. This eliminates the extra steps in the manufacturing process of creating and attaching the cap assemblies to the cable connector. Permanently tethering the cap assemblies, e.g., sealing caps, to the encapsulation member surrounding the rigid body helps ensure that utility workers will not drop or lose the cap assemblies, e.g., the sealing caps, once they are removed for a set screw installation. Additionally, the cap assemblies, e.g., sealing caps, of the present disclosure have a large pull tab that helps remove the cap assemblies by hand or with pliers. Hollowed-out sections of each of the cap assemblies make the cap assemblies of the present disclosure lighter and more flexible, which also helps with ease of installation as more rigid and bulky plugs tend to cause difficulty when installing. Further, the conductor ports of the cable connector are dimensioned to allow cable adapters, when inserted, to enter the rigid body of the cable connector by a predetermined distance, which limits and possibly prevents environmental elements from coming

into the rigid body via the conductor or cable ports.

[0005] In an exemplary embodiment, the submersible electrical cable connector for connecting a plurality of cable ends includes a connector body and an encapsulation overmold covering the connector body. The encapsulation overmold may also be referred to herein as the encapsulation member. The connector body has a front side and an upper side. The front side includes a plurality of cable receiving apertures, and the upper side includes a plurality of fastener receiving apertures in communication with and generally perpendicularly oriented to the cable receiving apertures. The encapsulation overmold includes integrally molded spaced apart tubular cable ports extending from the front side of the connector body and in communication with the cable receiving apertures. The encapsulation overmold also includes integrally molded spaced apart tubular fastener ports extending from the upper side of the connector body and in communication with the fastener receiving apertures. Tethered sealing caps are monolithically formed with the encapsulation overmold and each tethered sealing cap is configured to cap one of the fastener ports.

[0006] In another exemplary embodiment, the submersible electrical cable connector according to the present disclosure includes a connector body, an encapsulation overmold covering the connector body, and one or more cable adapters. The connector body has a front side and an upper side. The front side includes a plurality of cable receiving apertures, and the upper side includes a plurality of fastener receiving apertures in communication with and generally perpendicularly oriented to the cable receiving apertures. The encapsulation overmold includes spaced apart tubular cable ports extending from the front side of the connector body and in communication with the cable receiving apertures. The encapsulation overmold also includes spaced apart tubular fastener ports extending from the upper side of the connector body and in communication with the fastener receiving apertures. The cable adapters are shaped and sized for insertion into the cable ports and each cable port includes a length allowing respective one of the cable adapters to extend into the interior cavity of the connector body by a predetermined distance.

[0007] In another exemplary embodiment, the cable connector according to the present disclosure includes a connector body, an encapsulation member and a plurality of cap assemblies. The connector body has a first side and a second side. The first side includes a plurality of cable cavities, and the second side includes a plurality of fastener receiving apertures. In this configuration, one cable cavity is in communication with one fastener receiving aperture. The encapsulation member covers, encases or surrounds the connector body. The encapsulation member includes a plurality of cable ports and a plurality of fastener ports. Each cable port has a hollow center portion, and extends from the first side of the connector body such that the hollow center portion of the cable port is in communication with one of the cable cavities. Each fastener port has a hollow center portion defining an interior surface of the fastener port. Each fastener port extends from the second side of the connector body such that the hollow center portion of the fastener port is in communication with one of the fastener receiving apertures. Each cap assembly has a tether with a seal member at a first end. A second end of the tether is formed, e.g., monolithically formed, into the encapsulation member. The seal member of each cap assembly is preferably configured to be inserted into the hollow center portion of one of the fastener ports. The cable connector according to the present disclosure may also include a plurality of cable adapters. Each cable adapter is configured to receive a range of conductor sizes, and is shaped and sized for insertion into the hollow center portion of one of the cable ports. When one of the cable adapters is inserted into the hollow center portion of one of the cable ports, the cable adapter extends at least partially into one of the cable cavities.

[0008] In another exemplary embodiment, the cable connector according to the present disclosure includes a connector body, an encapsulation member, a plurality of cable ports, a plurality of fastener ports and a plurality of cap assemblies. The connector body has a first side and a second side. The first side includes a plurality of cable cavities and the second side includes a plurality of fastener receiving apertures, where one cable cavity is in communication with one fastener

receiving aperture. The encapsulation member surrounds the connector body and provides access to each of the plurality of cable cavities and each of the plurality of fastener receiving apertures. Each cable port has a hollow center portion, and extends from the first side of the connector body so that the hollow center portion of the cable port is in communication with one of the cable cavities. Each fastener port has a hollow center portion, and extends from the second side of the connector body so that the hollow center portion of the fastener port is in communication with one of the fastener receiving apertures. Each cap assembly includes a tether having a seal member at a first end. A second end of the tether is attached to or formed into, e.g., monolithically formed into, the encapsulation member. The seal member is configured to be at least partially inserted into the hollow center portion of one of the fastener ports.

[0009] In another exemplary embodiment, the cable connector according to the present disclosure includes a connector body, an encapsulation member and a plurality of cap assemblies. In this embodiment, the connector body has a first side and a second side. The first side includes a plurality of cable cavities and the second side includes a plurality of fastener receiving apertures, where one cable cavity is in communication with one fastener receiving aperture. The encapsulation member covers the connector body, and includes a plurality of cable ports and a plurality of fastener ports. The cable ports extend from the first side of the connector body. Each cable port has a hollow center portion, and one end integrally or monolithically molded into the encapsulation member so that the hollow center portion of the cable port is in communication with one of the cable cavities. The fastener ports extend from the second side of the connector body. Each fastener port has a hollow center portion, and one end integrally or monolithically molded into the encapsulation member so that the hollow center portion of the fastener port is in communication with one of the fastener receiving apertures. Each cap assembly includes a tether with a seal member at a first end. A second end of the tether is formed, e.g., monolithically formed, into the encapsulation member. The seal member is configured to be at least partially inserted into the hollow center portion of one of the fastener ports.

[0010] In the embodiments contemplated by the present disclosure, preferably, each cable port is integrally or monolithically molded into the encapsulation member, and each fastener port is integrally or monolithically molded into the encapsulation member.

[0011] In the embodiments contemplated by the present disclosure, each of the fastener receiving apertures may be at least partially threaded apertures so that a fastener can be threaded at least partially into each of the plurality of fastener receiving apertures.

[0012] In the embodiments contemplated by the present disclosure, the seal member for each of the cap assemblies may include at least one rib extending around an exterior surface of the seal member and outwardly from the exterior surface of the seal member so that when the seal member is inserted into the hollow center portion of one of the fastener ports the at least one rib contacts the interior surface of the fastener port.

[0013] In the embodiments contemplated by the present disclosure, each cable ports may include at least one supporting rib extending from an exterior surface of each of the cable port, and a portion of the at least one supporting rib may be formed, e.g., monolithically formed, into the encapsulation member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The figures depict embodiments for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures illustrated herein may be employed without departing from the principles described herein, wherein:

[0015] FIG. 1 is a perspective view of a submersible electrical cable connector according to the present disclosure;

[0016] FIG. 2 is a perspective view of the electrical cable connector of FIG. 1, illustrating cable adapters in accordance with the present disclosure immediately prior to being inserted into or after having been removed from wire way bosses of the electrical cable connector;

[0017] FIG. 3 is an exploded perspective view in partial cross-section of the electrical cable connector of FIG. 2, illustrating an exposed portion of a connector body after a portion of an encapsulation overmold has been removed from the connector body;

[0018] FIG. 4 is a cross-sectional view of the electrical cable connector of FIG. 1 taken along the lines 4-4, illustrating a relationship between the cable adapter and the wire way boss;

[0019] FIG. 5 is the cross-sectional view of the electrical cable connector of FIG. 4, illustrating an electrical conductor inserted in the connector body of the electrical cable connector;

[0020] FIG. 6a is a perspective view of an exemplary embodiment of a cable adapter according to the present disclosure;

[0021] FIG. 6b is a side elevation view of the cable adapter of FIG. 6a;

[0022] FIG. 6c is an end elevation view of the cable adapter of FIG. 6a;

[0023] FIG. 7a is a top perspective view of the sealing cap of the electrical cable connector of FIG. 1;

[0024] FIG. 7b is a bottom perspective view of the sealing cap of FIG. 7a;

[0025] FIG. 7c is a side elevation view of the sealing cap of FIG. 7a;

[0026] FIG. 7d is a cross-sectional view of the sealing cap of FIG. 7c taken along the line 7d-7d;

[0027] FIG. 8 is perspective view of another exemplary embodiment of the electrical cable connector according to the present disclosure;

[0028] FIG. 9 is a perspective view of the electrical cable connector of FIG. 8, illustrating cable adapters in accordance with the present disclosure immediately prior to being inserted into or after having been removed from wire way bosses of the electrical cable connector;

[0029] FIG. 10 is an exploded perspective view in partial cross-section of the electrical cable connector of FIG. 9, illustrating an exposed portion of a connector body after a portion of an encapsulation overmold has been removed from the connector body;

[0030] FIG. 11 is a cross-sectional view of the electrical cable connector of FIG. 8 taken along the lines 11-11, illustrating a relationship between the cable adapter and the wire way boss;

[0031] FIG. 12 is the cross-sectional view of the electrical cable connector of FIG. 11, illustrating an electrical conductor of a first size inserted in the connector body of the electrical cable connector;

[0032] FIG. 13 is the cross-sectional view of the electrical cable connector of FIG. 11, illustrating an electrical conductor of a second size inserted in the connector body of the electrical cable connector;

[0033] FIG. 14a is a top perspective view of the sealing cap of the electrical connector of FIG. 8;

[0034] FIG. 14b is a bottom perspective view of the sealing cap of FIG. 14a;

[0035] FIG. 14c is a side elevation view of the sealing cap of FIG. 14a; and

[0036] FIG. 14d is a cross-sectional view of the sealing cap of FIG. 14c taken along the line 14d-14d.

DETAILED DESCRIPTION

[0037] The present disclosure provides exemplary embodiments of a submersible electrical cable connector with tethered sealing caps used to cover openings in the submersible electrical cable connector. The submersible electrical cable connector according to the present disclosure is to be used to make quick and easy tap connections, terminations, and in-line splices of one or more conductors in a variety of configurations. Non-limiting examples of the one or more conductors include a single electrical conductor wire, a bundle of electrical conductor wires, laminated strips of conductors, any combination thereof or any other aluminum, copper-clad aluminum, or copper

conductor. For ease of description, the submersible electrical cable connector may also be referred to as the “cable connector” in the singular and as the “cable connectors” in the plural. Also, for ease of description, the sealing cap may also be referred to as the “cap” in the singular and as the “caps” in the plural. The specification and drawings are to be regarded in an illustrative sense rather than a restrictive sense. Various modifications may be made thereto without departing from the spirit and scope as set forth in the following claims.

[0038] Referring now to FIGS. 1-5 and 6a-6c, an exemplary embodiment of a cable connector **10** according to the present disclosure is shown. In this exemplary embodiment, the cable connector **10** includes a connector body **12** that is encased or covered in a layer of insulating material **14**, as seen in FIG. 3. In the exemplary embodiment shown, the connector body **12** is a generally rectangular body. However, the connector body may be square in shape, cylindrical in shape or any other suitable shape to make the electrical connections described herein. The connector body **12** is preferably formed of electrically conductive metallic material, such as aluminum, aluminum alloy, copper or brass. The connector body **12** includes one or more spaced apart wire cavities **13** arranged along one side of the connector body **12** and one or more spaced apart apertures **15** arranged along an adjacent side of the connector body **12**, as shown in FIG. 3. In the exemplary embodiment shown in FIG. 3, each aperture **15** is oriented at right angle relative to a respective wire cavity **13**. Each aperture **15** intersects the wire cavity **13**, as seen in FIG. 4, so that the wire way cavity **13** of the connector body **12** is in communication with the respective aperture **15**. Preferably, each wire cavity **13** and aperture **15** are blind holes, i.e., they extend inwardly into the connector body **12**, but do not extend fully through the connector body **12**.

[0039] The layer of insulating material **14** may be an injection molded thermoplastic or elastomeric compound that is formed to substantially surround or overmold the connector body **12**. For ease of description, the layer of insulating material **14** may also be referred to herein as the encapsulation member **14**. The encapsulation member **14** is formed or molded to include one or more spaced apart tubular and generally hollow ports **16**, each of which may also be referred to herein as a wire way boss. Each wire way boss **16** extends outwardly from the same side of the connector body **12** as the wire cavities **13** so that the wire cavities **13** are in communication with the hollow portion of the wire way bosses **16**. The wire way bosses **16** provide access to the wire cavities **13** so that one or more conductors **60** can pass through the wire way bosses **16** into the wire cavity **13**, as shown in FIG. 5. The encapsulation member **14** also includes one or more spaced apart tubular and generally hollow ports **18**, each of which may also be referred to herein as a fastener boss. Each fastener boss **18** extends outwardly from the same side of the connector body **12** as the aperture **15** so that the apertures **15** are in communication with the hollow portion **19** of the fastener bosses **18**. The fastener bosses **18** provide access to the apertures **15** so that one or more fasteners **58**, e.g., set screws, in the connector body **12** can be accessed by utility workers with conventional tools, or so that one or more fasteners **58** can be pass through the fastener bosses **18** into the apertures **15**, as shown in FIGS. 3-5. As it should be apparent from FIG. 5, each fastener **58** can then be tightened to secure a conductor **60** extending in respective one of the wire cavities **13** to the connector body **12**. In the exemplary embodiment shown, like each wire cavity **13** and aperture **15** pair, the wire way bosses **16** are oriented at right angle to the fastener bosses **18**.

[0040] Continuing to refer to FIGS. 3-5, the encapsulation member **14** is formed or molded to further include one or more cap assemblies **20**. Each cap assembly **20** includes a seal member **22** and a tether **24**, seen in FIG. 3. For ease of description, the cap assemblies **20** may also be referred to herein as the “cap” in the singular and the “caps” in the plural. The tether **24** of the cap **20** is preferably monolithically or integrally formed into the encapsulation member **14** during, for example, the injection molding process so that the tether **24** is permanently attached to the encapsulation member **14**. The seal members **22** of the caps **20**, which are described in more detail below, are configured to cover or cap the hollow openings **19** of the fastener bosses **18**, as shown in FIGS. 1-5.

[0041] As mentioned, the process that forms the encapsulation member **14**, e.g., an injection molding process, also forms the caps **20** as one monolithically or integrally formed unit. The tether **24** of each cap **20** is a flexible tether that attaches each seal member **22** of the cap **20** to the encapsulation member **14**, as shown in FIGS. 1-5. In the exemplary embodiments of FIGS. 1-5 the tether **24** is an elongated member having a first end connected to the encapsulation member **14** at a side of the connector body **12** and a second end connected to the seal member **22** of the cap **20**. However, it is contemplated that the first end of the tether **24** may be monolithically or integrally formed into the encapsulation member **14** at any other location along the connector **10** such that the seal member **22** of the cap **20** can be easily removed and inserted back into the hollow opening **19** of the fastener boss **18**. It is further contemplated that the junction between the first end of the tether **24** and the encapsulation member **14** may be thicker than the rest of the tether **24** to help better withstand repeated bending and flexing of the tether **24**.

[0042] Turning now to FIGS. 7a-7d, each seal member **22** of each cap **20** is configured to provide an environmental seal for each respective opening **19** of the fastener bosses **18** before and after a respective fastener **58** has been secured the conductor **62** in the wire way cavity **13** of the connector body **12**. Each seal member **22** includes a cylinder **26** having a proximal end portion **26a** and a distal end portion **26b**. The cylinder **26** may be a solid cylinder or a substantially hollow or hollowed out cylinder that has an outer diameter sufficient to fit within the opening of the fastener bosses **18** and environmentally seal the opening of the fastener bosses **18**. In the exemplary embodiment shown, the cylinder **26** is a substantially hollow cylinder. The proximal end portion **26a** includes a flanged portion **28** that has a larger diameter than a diameter of the cylinder **26** and that extends around a periphery of the proximal end portion **26a**. The outer diameter of the flange portion **28** is preferably larger than the inner diameter of the opening of the fastener bosses **18** so that the flange portion **28** remains outside the opening of the fastener bosses **18**. As best seen in FIGS. 4 and 7d, the second end of the tether **24** is connected to the flanged portion **28**. The internal cavity within the substantially hollow cylinder **26** does not extend through the cylinder **26**. Instead, the internal cavity is closed off roughly halfway along the longitudinal extent of the cylinder **26** by a partition **30**, which forms an upper cavity **32** and a lower cavity **34**, as shown in FIG. 7d. The upper cavity **32** includes a gripping member or tab **36** extending from the partition **30** within the upper cavity **32** beyond the flanged portion **28**. The tab **36** facilitates grasping, e.g., manual grasping or tool aided grasping, of each seal member **22** of each cap **20** to remove the seal member **22** from or insert the seal member into a fastener boss **18**. As shown in FIGS. 3, 7b and 7d, the lower cavity **34** includes a pair of cross members **38** that transversely extend from the partition **30** within the lower cavity **34**. The lower cavity **34** is provided for weight reduction and cost savings, and the cross members **38** are provided for added structural support. An exterior surface of the distal end portion **26b** further includes one or more outwardly extending peripheral ribs **40** that are configured to provide an interference fit connection or snap fit connection between each seal member **22** of each cap **20** and respective one of the fastener bosses **18**.

[0043] With reference to FIGS. 2-4 and 6, the cable connector **10** may include one or more removable cable adapters **42**. The cable adapters **42** are preferably elastomeric members, which may also be referred to as wire way rockets, that may be formed via the same process, e.g., injection molding process, used to form the encapsulation member **14**. Each cable adapter **42** is formed to have a predefined shape and size for insertion into one of the wire way bosses **16** to help seal a range of conductor sizes inserted into the wire way bosses **16** and/or to provide an environmental seal for the wire cavities **13** in the connector body **12** that do not have a conductor inserted therein. As best seen in FIGS. 6a-6c, the cable adapter **42** is generally conically shaped and includes a radially extending base portion **44**, a longitudinally extending tubular lower portion **45**, a longitudinally extending tubular sidewall **46**, and a closed end portion **48**. A diameter of the base portion **44** is larger than a diameter of the lower portion **45**, which has a larger diameter than a diameter of the sidewall **46**, such that when one of the cable adapters **42** is inserted into one of the

wire way bosses **16**, the base portion **44** is configured to abut an outer edge of the wire way boss **16**, as can be seen in FIGS. **1** and **4**.

[0044] Continuing to refer to FIGS. **6a-6c**, the sidewall **46** extends between the lower portion **45** and the closed end portion **48**. The diameter of the sidewall **46** is stepped or sectionally tapered down along the length of the sidewall **46** to allow the utility worker to sever one or more steps or sections of the sidewall **46** along the length thereof to accommodate a correspondingly shaped or sized conductor to be inserted through the cable adapter **42** into the wire cavity **13**, as seen in FIG. **5**. Although in the exemplary embodiment of FIGS. **6a** and **6b** the sidewall **46** includes three steps **90**, **92** and **94** having diminished diameters, the sidewall **46** may be formed to include more steps, each having a smaller diameter than a preceding step. However, in other embodiments, the diameter of the sidewall **46** may be a smooth taper instead of a stepped taper.

[0045] The sidewall **46** of the cable adapter **42** may include markings imprinted or molded thereon identifying the proper conductor range or diameter accommodated by the corresponding severed steps or sections thereof, such as the markings shown in FIGS. **6a-6c**. For instance, the utility worker would sever or cut off the closed end portion **48**, which has a smallest diameter opening in the cable adapter **42** to insert a #12 AWG to #8 AWG electrical conductor through the cable adapter **42** into the wire cavity **13**. To insert a #6 AWG to #2 AWG electrical conductor through the cable adapter **42**, the utility worker would sever two steps or sections, namely the closed end portion **48** and the adjacent step or section **94**. The interior surface of the lower portion **45** and/or sidewall **46**, shown in FIG. **4**, may include radially extending protrusions **49** that extend into the hollow portion of the cable adapter **42** to enhance the grip of the cable adapter **42** on a conductor extending therethrough. The lower portion **45** of each cable adapter **42** may include outwardly extending peripheral ribs **50**, seen in FIGS. **6a** and **6b**, that are designed to limit or possibly prevent water from passing through the wire boss **16** and entering the wire cavity **13**, as will be discussed in more detail below.

[0046] As noted above, each wire cavity **13** in the connector body **12** is configured to receive an electrical conductor **60**, seen in FIG. **5**, that can be inserted through a cable adapter **42** positioned withing a wire way boss **16**. In the embodiment shown, the conductor **60** is an insulated conductor having an electrical conductor (aka a wire) with an insulating jacket. The conductor **60** can then be extended into the wire cavity **13** of the connector body **12**. Prior to insertion of the conductor **60** into the wire way boss **16**, a utility worker may remove a portion of the insulation surrounding the wire **62** of the conductor **60** so that a predetermined length of wire is exposed. In turn, the utility worker can insert a tool, e.g., an Allen wrench or screwdriver, into the fastener boss **18** and torque down the fastener **58** to secure the exposed wires **62** in the wire way cavity **22** to the connector body **12**.

[0047] Referring again to FIG. **4**, each wire way boss **16** extends from a side of the connector body **12** by a sufficient length to permit the cable adapter **42** inserted into wire way boss **16** to enter the wire cavity **13** of the connector body **12**. More particularly, each of the wire way boss **16** has the length allowing the cable adapter **42** inserted therethrough to extend into the wire cavity **13** of the connector body **12** at least to a plane A, seen in FIG. **4**, to help minimize flexing of the cable adapter **42** when a wire **62** is secured to the connector body **12** by torquing the fastener **58**. Preferably, the cable adapter **42** is not extended into the wire cavity beyond a plane B so as to avoid the cable adapter **42** interfering with the fastener **58** contacting the wire **62** when securing the wire **62** to the connector body **12**.

[0048] An exterior surface of each wire way boss **16** may include a plurality of longitudinally extending supporting ribs **52** radially disposed around the periphery of each wire way boss **16**, as shown for example in FIGS. **1**, **4** and **5**. The supporting ribs **52** provide additional structural support and rigidity for the wire way bosses **16**. Additionally, as can be seen in FIGS. **2** and **3**, a portion of the interior surface of each wire way boss **16** may include outwardly and radially extending grooves **54** that are configured to mate with or engage the peripheral ribs **50** of an

inserted cable adapter **42** to limit or possibly prevent water from passing through the wire way boss **16** and entering the wire cavity **13** of the connector body **12**.

[0049] Referring now to FIGS. **8-13**, another exemplary embodiment of a cable connector **100** according to the present disclosure is shown. In this exemplary embodiment, the cable connector **100** includes a connector body **112** that is encased or covered in a layer of insulating material **114**, as seen in FIG. **10**. In the exemplary embodiment shown, the connector body **112** is a generally rectangular body. However, the connector body **112** may be square in shape, cylindrical in shape or any other suitable shape to make the electrical connections described herein. The connector body **112** is preferably formed of electrically conductive metallic material, such as aluminum, aluminum alloy, copper or brass. The connector body **112** includes one or more spaced apart wire cavities **113** arranged along one side of the connector body **112** and one or more spaced apart apertures **115** arranged along an adjacent side of the connector body **112**, as shown in FIG. **10**. In the exemplary embodiment shown in FIG. **10**, each aperture **115** is oriented at right angle relative to a respective wire cavity **113**. Each aperture **115** intersects the wire cavity **113**, as seen in FIG. **11**, so that the wire way cavity **113** of the connector body **112** is in communication with the respective aperture **115**. Preferably, each wire cavity **113** and aperture **115** are blind holes, i.e., they extend inwardly into the connector body **112**, but do not extend fully through the connector body **112**.

[0050] The layer of insulating material **114** may be an injection molded thermoplastic or elastomeric compound that is formed to substantially surround or overmold the connector body **112**. For ease of description, the layer of insulating material **114** may also be referred to herein as the encapsulation member **114**. The encapsulation member **114** is formed or molded to include one or more spaced apart tubular and generally hollow ports **116**, each of which may also be referred to herein as a “wire way boss.” Each wire way boss **116** extends outwardly from the same side of the connector body **112** as the wire cavities **113** so that the wire cavities are in communication with the hollow portion of the wire way bosses **116**. The wire way bosses **116** provide access to the wire cavities **113** so that one or more conductors, e.g., conductors **160** and **162**, can pass through the wire way bosses **116** into the wire cavity **113**, as shown in FIGS. **12** and **13**. The encapsulation member **114** also includes one or more spaced apart tubular and generally hollow ports **118**, each of which may also be referred to herein as a “fastener boss.” Each fastener boss **118** extends outwardly from the same side of the connector body **112** as the aperture **115**, seen in FIG. **10**, so that the apertures are in communication with the hollow portion of the fastener bosses **118**. The fastener bosses **118** provide access to the apertures **115** so that one or more fasteners **158**, e.g., set screws, in the connector body **112** can be accessed by utility workers with conventional tools, or so that one or more fasteners **158** can be pass through the fastener bosses **118** into the apertures **115**, as shown in FIGS. **10-13**. As it should be apparent from FIGS. **12** and **13**, each fastener **158** can then be tightened to secure a wire extending in respective one of the wire cavities **113** to the connector body **112**. In the exemplary embodiment shown, like each wire cavity **113** and aperture **115** pair, the wire way bosses **116** are oriented at right angles relative to the fastener bosses **118**.

[0051] Continuing to refer to FIGS. **10-13**, the encapsulation member **114** is formed or molded to further include one or more cap assemblies **120**. Each cap assembly **120** includes a seal member **122** and a tether **124**. For ease of description, the cap assemblies **120** may also be referred to herein as the “cap” in the singular and the “caps” in the plural. The tether **124** of the cap **120** is preferably monolithically or integrally formed into the encapsulation member **114** during, for example, the injection molding process so that the tether is permanently attached to the encapsulation member **114**. The seal members **122** of the caps **120**, which are described in more detail below, are configured to cover or cap the hollow openings of the fastener bosses **118**, as shown in FIGS. **11-13**.

[0052] As mentioned, the process that forms the encapsulation member **114**, e.g., an injection molding process, also forms the caps **120** as one monolithically or integrally formed unit. The tether **124** of each cap **120** is a flexible tether that attaches each seal member **122** of the cap **120** to

the encapsulation member **114**, as shown in FIGS. **10-13**. In the exemplary embodiments of FIGS. **10-13** the tether **124** is an elongated member having a first end attached to the encapsulation member **114** at a side of the connector body **112** and a second end connected to the seal member **122** of the cap **120**. However, it is contemplated that the first end of the tether **124** may be monolithically or integrally formed into the encapsulation member **114** at any other location along the connector **100** such that the seal member **122** of the cap **120** can be easily removed and inserted back into the hollow opening **119** of the fastener boss **118**. It is further contemplated that the junction between the first end of the tether **124** and the encapsulation member **114** may be thicker than the rest of the tether **124** to help better withstand repeated bending and flexing of the tether **124**.

[0053] Turning now to FIGS. **14a-14d**, each seal member **122** of each cap **120** is configured to provide an environmental seal for each respective opening of the fastener bosses **118** before and after a respective fastener **158** has been secured the conductor, conductor **160** and **162**, in the wire way cavity **113** of the connector body **112**. Each seal member **122** includes an insert portion **126** and a flange portion **128**. In this exemplary embodiment, the insert portion **126** has a proximal end portion **126a** and a distal end portion **126b**. The insert portion **126** may be a solid member or a substantially hollow or a hollowed out member that has an outer perimeter, e.g., diameter, sufficient to fit within the opening of the fastener bosses **118** and environmentally seal the opening of the fastener bosses **118**. The insert portion **126** is shaped to fit within the opening **119** of the fastener bosses **118**. For example, the insert portion **126** may have a cylindrical, square, rectangular, triangular or other shape to fit within the opening **119** of the fastener bosses. In the embodiment shown in FIGS. **14a-14d**, the insert portion **126** is a cylindrical member that is substantially hollow having one or more internal cavities **134**. The one or more internal cavities **134** within the substantially hollow insert portion **126** may or may not extend through the entire length of the insert portion **126**. In the exemplary embodiment shown, the one or more internal cavities **134** within the insert portion **126** extend through the entire length of the insert portion **126**. As shown in FIGS. **10, 14b** and **14d**, the substantially hollow insert portion **126** includes a pair of cross members **138** that transversely extend from one side of the insert portion **126** to an opposite side of the insert portion **126** and form the one or more cavities **134**. The one or more cavities **34** are provided for weight reduction and cost savings, and the cross members **138** are provided as added structural support for the insert portion **126**. An exterior surface of the distal end portion **126b** of the insert portion **126** may include one or more outwardly extending peripheral ribs **140** that are configured to provide an interference fit connection or snap fit connection between each seal member **122** of each cap **120** and a respective one of the fastener bosses **118**.

[0054] Continuing to refer to FIGS. **14a-14d**, the flanged portion **128** of the seal member **122** may be integral with or monolithically formed to the proximal end portion **126a** of the insert portion **126**, or the flanged portion **128** may be secured to the proximal end portion **126a** of the insert portion **126** using, for example, adhesives or welds, e.g., sonic welds. Preferably, the flanged portion **128** has a larger perimeter, e.g., diameter, than an outer perimeter, e.g., outer diameter, of the insert portion **126**. The outer perimeter, e.g., the outer diameter, of the flange portion **128** is preferably larger than the inner diameter of the opening of the fastener bosses **118** so that the flange portion **128** remains outside the opening of the fastener bosses **118**. The flanged portion **128** of the seal member **122** may be integral with or monolithically formed to the second end **124a** of the tether **124**, or the flanged portion **128** of the seal member **122** may be secured to the second end of the tether **124** using, for example, adhesives or welds, e.g., sonic welds. As best seen in FIGS. **11** and **14d**, the flanged portion **128** of the seal member **122** is integral with or monolithically formed to the second end **124a** of the tether **124**. An upper surface **128a** of the flange member **128** may include a gripping member or tab **136** extending therefrom. The tab **136** facilitates grasping, e.g., manual grasping or tool aided grasping, of each seal member **122** of each cap **120** when removing the seal member **22** from or inserting the seal member into a fastener boss **118**.

[0055] With reference to FIGS. **6a**, **6b**, **6c** and **8-10**, the cable connector **100** may also include one or more removable cable adapters **42** having a base portion **44**, a longitudinally extending tubular lower portion **45**, a longitudinally extending tubular sidewall **46**, and a closed end portion **48**. The cable adapters **42** are described above and for ease of description and clarity will not be repeated here. It is noted however, that the sidewall **46** extends between the lower portion **45** and the closed end portion **48**, and the diameter of the sidewall **46** is stepped or sectionally tapered down along the length of the sidewall **46** to allow the utility worker to sever one or more steps or sections of the sidewall **46** along the length thereof to accommodate a correspondingly shaped or sized conductor to be inserted through the cable adapter **42** into the wire cavity **13**, as seen in FIGS. **12** and **13**.

[0056] As noted above, each wire cavity **113** in the connector body **112** is configured to receive an electrical conductor, e.g., conductor **160**, seen in FIG. **12**, and conductor **162**, seen in FIG. **13**. The conductors **160** and **162** are different gauge conductors. Typically, the electrical conductors are insulated conductors having an electrical wire surrounded by insulating jacket. The electrical conductor can be inserted through the cable adapter **42** positioned within a wire way boss **116** into the wire cavity **113** of the connector body **112**. Prior to insertion of the conductor **160** through the cable adapter **42** positioned within a wire way boss **116**, a utility worker may remove a portion of the insulation surrounding the wire **164** of the conductor **160** or the wire **166** of the conductor **162** so that a predetermined length of wire is exposed. Once the wire is positioned the wire cavity **113**, the utility worker can insert a tool, e.g., an Allen wrench or screwdriver, into the fastener boss **118** and torque down the fastener **158** to secure the wire in the wire cavity **113** to the connector body **112**.

[0057] Referring again to FIG. **11**, each wire way boss **116** extends from a side of the connector body **12** by a sufficient length to permit the cable adapter **42** inserted into wire way boss **116** to enter the wire cavity **113** of the connector body **112**. More particularly, each wire way boss **116** has a length allowing the cable adapter **42** inserted therethrough to extend into the wire cavity **113** of the connector body **112** at least to a plane A, seen in FIG. **11**, to help minimize flexing of the cable adapter **42** when a wire **164**, **166** is secured to the connector body **112** by torquing the fastener **158**. Preferably, the cable adapter **42** is not extended into the wire cavity beyond a plane B so as to avoid the cable adapter **42** interfering with the fastener **158** contacting the wire, e.g., wire **164**, or **166**, when securing the wire to the connector body **112**.

[0058] Referring again to FIGS. **8-10**, an exterior surface of each wire way boss **116** may include a plurality of longitudinally extending supporting ribs **152** radially disposed around the periphery of each wire way boss **116**. Each supporting rib **152** is configured and dimensioned to provide additional structural support and rigidity for the wire way bosses **116**. In the embodiment of FIGS. **8-10**, the plurality of supporting ribs **152** may be integral with or monolithically formed into the encapsulation member **114** and/or the wire way bosses **116**, or the plurality of supporting ribs **152** may be secured to the encapsulation member **114** and/or the wire way bosses **116** using, for example, adhesives or welds, e.g., sonic welds. Additionally, as can be seen in FIGS. **8** and **9**, a portion of the interior surface of each wire way boss **116** may include one or more outwardly and radially extending grooves **154** that are configured to mate with or engage the peripheral ribs **50** of an inserted cable adapter **42** to limit or possibly prevent water from passing through the wire way boss **116** and entering the wire cavity **113** of the connector body **112**.

[0059] While illustrative embodiments of the present disclosure have been described and illustrated above, it should be understood that these are exemplary of the disclosure and are not to be considered as limiting. Additions, deletions, substitutions, and other modifications can be made without departing from the spirit or scope of the present disclosure. Accordingly, the present disclosure is not to be considered as limited by the foregoing description.

Claims

- 1.** A cable connector comprising: a connector body having a first side and a second side, the first side including a plurality of cable cavities and the second side including a plurality of fastener receiving apertures, where one cable cavity is in communication with one fastener receiving aperture; an encapsulation member covering the connector body, the encapsulation member including: a plurality of cable ports, each of the plurality of cable ports having a hollow center portion, each of the plurality of cable ports extend from the first side of the connector body such that the hollow center portion of the cable port is in communication with one of the plurality of cable cavities; a plurality of fastener ports, each of the plurality of fastener ports having a hollow center portion defining an interior surface of the fastener port, each of the plurality of fastener ports extend from the second side of the connector body such that the hollow center portion of the fastener port is in communication with one of the plurality of fastener receiving apertures; and a plurality of cap assemblies, each of the plurality of cap assemblies having a tether with a seal member at a first end of the tether and a second end of the tether formed into the encapsulation member, wherein the seal member of each of the plurality of cap assemblies is configured to be inserted into the hollow center portion of one of the plurality of fastener ports.
- 2.** The cable connector according to claim 1, further comprising a plurality of cable adapters, each of the plurality of cable adapters is configured to receive a range of conductor sizes and shaped and sized for insertion into the hollow center portion of one of the plurality of cable ports, wherein when one of the plurality of cable adapters is inserted into the hollow center portion of one of the plurality of cable ports, the one of the plurality of cable adapters extends at least partially into one of the plurality of cable cavities.
- 3.** The cable connector according to claim 1, wherein each of the plurality of cable ports is integrally molded into the encapsulation member.
- 4.** The cable connector according to claim 1, wherein each of the plurality of fastener ports is integrally molded into the encapsulation member.
- 5.** The cable connector according to claim 1, wherein each of the plurality of fastener receiving apertures are at least partially threaded apertures, and a fastener is threaded at least partially into each of the plurality of fastener receiving apertures.
- 6.** The cable connector according to claim 1, wherein the seal member for each of the plurality of cap assemblies includes at least one rib extending around an exterior surface of the seal member and outwardly from the exterior surface of the seal member such that when the seal member is inserted into the hollow center portion of one of the plurality of fastener ports the at least one rib contacts the interior surface of the one of the plurality of fastener ports.
- 7.** The cable connector according to claim 1, wherein each of the plurality of cable ports has an exterior surface, and each of the plurality of cable ports includes at least one supporting rib extending from the exterior surface of each of the plurality of cable ports, and wherein a portion of the at least one supporting rib is molded into the encapsulation member.
- 8.** A cable connector comprising: a connector body having a first side and a second side, the first side including a plurality of cable cavities and the second side including a plurality of fastener receiving apertures, where one cable cavity is in communication with one fastener receiving aperture; an encapsulation member surrounding the connector body and providing access to each of the plurality of cable cavities and each of the plurality of fastener receiving apertures; a plurality of cable ports, each of the plurality of cable ports having a hollow center portion, and each of the plurality of cable ports extending from the first side of the connector body such that the hollow center portion of one of the plurality of cable ports is in communication with one of the plurality of cable cavities; a plurality of fastener ports, each of the plurality of fastener ports having a hollow center portion, each of the plurality of fastener port extending from the second side of the connector body such that the hollow center portion of one of the plurality of fastener ports is in communication with one of the plurality of fastener receiving apertures; and a plurality of cap

assemblies, where each of the plurality of cap assemblies includes a tether having a seal member at a first end and a second end attached to or formed into the encapsulation member, the seal member being configured to be at least partially inserted into the hollow center portion of one of the plurality of fastener ports.

9. The cable connector according to claim 8, further comprising a plurality of cable adapters, each of the plurality of cable adapters is configured to receive a range of conductor sizes and is shaped and sized for insertion into the hollow center portion of one of the plurality of cable ports, wherein when one of the plurality of cable adapters is inserted into the hollow center portion of one of the plurality of cable ports, the one of the plurality of cable adapters extends at least partially into one of the plurality of cable cavities.

10. The cable connector according to claim 8, wherein each of the plurality of cable ports is integrally molded into the encapsulation member.

11. The cable connector according to claim 8, wherein each of the plurality of fastener ports is integrally molded into the encapsulation member.

12. The cable connector according to claim 8, wherein each of the plurality of fastener receiving apertures are at least partially threaded apertures, and a fastener is threaded at least partially into each of the plurality of fastener receiving apertures.

13. The cable connector according to claim 8, wherein the seal member for each of the plurality of cap assemblies includes at least one rib extending around an exterior surface of the seal member and outwardly from the exterior surface of the seal member such that when the seal member is inserted into the hollow center portion of one of the plurality of fastener ports the at least one rib contacts an interior surface of the one of the plurality of fastener ports fastener ports.

14. The cable connector according to claim 8, wherein each of the plurality of cable ports has an exterior surface, and each of the plurality of cable ports includes at least one supporting rib extending from the exterior surface of each of the plurality of cable ports, and wherein a portion of the at least one supporting rib is molded into the encapsulation member.

15. A cable connector comprising: a connector body having a first side and a second side, the first side including a plurality of cable cavities and the second side including a plurality of fastener receiving apertures, where one cable cavity is in communication with one fastener receiving aperture; an encapsulation member covering the connector body, the encapsulation member including: a plurality of cable ports extending from the first side of the connector body, each of the plurality of cable ports having a hollow center portion, and each of the plurality of cable ports having one end integrally molded into the encapsulation member such that the hollow center portion of one of the plurality of cable ports is in communication with one of the plurality of cable cavities; a plurality of fastener ports extending from the second side of the connector body, each of the plurality of fastener ports having a hollow center portion, each of the plurality of fastener port having one end integrally molded into the encapsulation member such that the hollow center portion of the fastener port is in communication with one of the plurality of fastener receiving apertures; and a plurality of cap assemblies, each of the plurality of cap assemblies having a tether with a seal member at a first end of the tether and a second end of the tether formed into the encapsulation member, the seal member being configured to be at least partially inserted into the hollow center portion of one of the plurality of fastener ports.

16. The cable connector according to claim 15, further comprising a plurality of cable adapters, each of the plurality of cable adapters is configured to receive a range of conductor sizes and shaped and sized for insertion into the hollow center portion of one of the plurality of cable ports, wherein when one of the plurality of cable adapters is inserted into the hollow center portion of one of the plurality of cable ports, the one of the plurality of cable adapters extends at least partially into one of the plurality of cable cavities.

17. The cable connector according to claim 15, wherein each of the plurality of fastener receiving apertures are at least partially threaded apertures, and a fastener is threaded at least partially into

each of the plurality of fastener receiving apertures.

18. The cable connector according to claim 15, wherein the seal member for each of the plurality of cap assemblies includes at least one rib extending around an exterior surface of the seal member and outwardly from the exterior surface of the seal member such that when the seal member is inserted into the hollow center portion of one of the plurality of fastener ports the at least one rib contacts an interior surface of the one of the plurality of fastener ports fastener ports.

19. The cable connector according to claim 15, wherein each of the plurality of cable ports has an exterior surface, and each of the plurality of cable ports includes at least one supporting rib extending from the exterior surface of each of the plurality of cable ports, and wherein a portion of the at least one supporting rib is molded into the encapsulation member.
